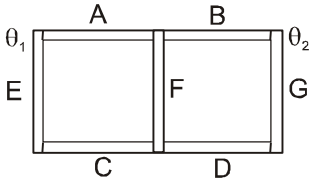
High Level Problems (HLP)

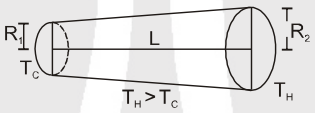
Marked Questions can be used as Revision Questions.

SUBJECTIVE QUESTIONS

1. Seven rods A, B, C, D, E, F and G are joined as shown in figure. All the rods have equal cross-sectional area A and length l. The thermal conductivities of the rods are $K_A = 2K_C = 3K_B = 6K_D = K_0$. The rod E is kept at a constant temperature θ_1 and the rod G is kept at a constant temperature $\theta_2(\theta_2 > \theta_1)$. (a) Show that the rod F has a uniform temperature $\theta = (3\theta_1 + \theta_2)/4$. (b) Find the rate of heat flow from the source which maintains the temperature θ_2.



2. Find the rate of heat flow through a cross-section of the rod shown in figure ($T_H > T_C$). Thermal conductivity of the material of the rod is K.



3. A solid aluminium sphere and a solid copper sphere of twice the radius of aluminium are heated to the same temperature and are allowed to cool under identical surrounding temperatures. Assume that the emissivity of both the sphere is the same. Find the initial ratio of (a) the rate of heat loss from the aluminium sphere to the rate of heat loss from the copper sphere and (b) the rate of fall of temperature of the aluminium sphere to the rate of fall of temperature of the copper sphere. The specific heat capacity of aluminium = 900 J/kg–°C and that of copper = 390 J/kg–°C. The density of copper = 3.4 times the density of aluminium.

4. A hot body placed in a surrounding of temperature T_0. Its temperature at t = 0 is T_1. The specific heat capacity of the body is s and its mass is m. Assuming Newton's law of cooling to be valid, find (a) the maximum heat that the body can lose and (b) the time starting from t = 0 in which it will lose 50% of this maximum heat. (Answer in terms of k)

5. Find the total time elapsed for a hollow copper sphere of inner radius 3 cm outer radius 6 cm, density $\rho = 9 \times 10^3$ kg/m^3, specific heat $s = 4 \times 10^3$ J/kg K and emissivity e = 0.4 to cool from 727°C to 227°C when the surrounding temperature is 0 K.(for inner surface e = 1 Stefan's constant $\sigma = 5.6 \times 10^{-8}$ W/m^2 K^4)

6. A metal block of heat capacity 90 J/°C placed in a room at 25°C is heated electrically. The heater is switched off when the temperature reaches 35°C. The temperature of the block rises at the rate of 2°C/s just after the heater is switched on and falls at the rate of 0.2 °C/s just after the heater is switched off. Assume Newton's law of cooling to hold. (a) Find the power of the heater. (b) Find the power radiated by the block just after the heater is switched off. (c) Find the power radiated by the block when the temperature of the block is 30°C. (d) Assuming that the power radiated at 30°C respresents the average value in the heating process, find the time for which the heater was kept on.

7. A hollow tube has a length l, inner radius R_1 and outer radius R_2. The material has thermal conductivity K. Find rate of heat flowing through the walls of the tube if the flat ends are maintained at temperatures T_1 and $T_2(T_2 > T_1)$.

8. Calculate thermal conductance for radial flow of an annular cylinder of length ℓ and inner and outer radius r_1 and r_2. Assume that thermal conductivity of the material is K



9. Calculate thermal conductance for radial flow of a spherical shell of inner and outer radius r_1 and r_2. Assume that thermal conductivity of the material is K

10. A metallic cylindrical vessel whose inner and outer radii are r_1 and r_2 is filled with ice at 0ºC. The mass of the ice in the cylinder is m. Circular portions of the cylinder is sealed with completely adiabatic walls. The vessel is kept in air. Temperature of the air is 50ºC. Find time elapsed for the ice to melt completely. (Thermal conductivity of the cylinder is k, its length is ℓ. Latent heat of fusion is L)

11. A uniform cylinder of length L and thermal conductivity k is placed on a metal plate of the same area S of mass m and infinite conductivity. The specific heat of the plate is c. The top of the cylinder is maintained at T_0. Find the time required for the temperature of the plate to rise from T_1 to T_2 ($T_1 < T_2 < T_0$).

12. Assume that the total surface area of a human body is 1.6 m^2 and that it radiates like an ideal radiator. Calculate the amount of energy radiated per second by the body if the body temperature is 37°C. Stefan constant σ is 6.0×10^{-8} $W/m^2–K^4$. ($31^4 = 923521$)

13. The surface of a household radiator has an emissivity of 0.55 and an area of 1.5 m^2.
(a) At what rate is radiation emitted by the radiator when its temperature is 50°C?
(b) At what rate is the radiation absorbed by the radiator when the walls of the room are at 22°C? (c) What is the net rate of radiation from the radiator? (stefan constant $\sigma = 6 \times 10^{-8}$ $W/m^2 – K^4$)

14. A man, the surface area of whose skin is $2m^2$, is sitting in a room where the air temperature is 20°C. If the skin temperature is 28°C. Find the net rate at which his body loses heat.
[Take the emissivity of skin 0.97 and stephen's constant = 5.67×10^{-8} $W/m^2 – K^4$

15. An electric heater is used in a room of total wall area 137 m^2 to maintain a temperature of 20°C inside it when outside temperature is –10°C. The walls have three different layers of materials. The innermost layer is of wood of thickness 2.5 cm, the middle layer is of cement of thickness 1.0 cm and the outermost layer is of brick of thickness 25 cm. Find the power of the electric heater :-
(Assume that there is no heat loss through the floor and the ceiling. The thermal conductivities of wood, cement and brick are 0.125, 1.5 and 1.0 W/m C° respectively).

16. A rod of length ℓ with thermally insulated lateral surface consists of material whose heat conductivity coefficient varies with temperature as $K = \alpha / T$, where α is a constant. The ends of the rod are kept at temperatures T_1 and T_2. Find the function T(x), where x is the distance from the end whose temperature is T_1 and the heat flow density,

17. Two chunks of metal with heat capacities C_1 and C_2, are interconnected by a rod of length ℓ and cross-sectional area S and fairly low heat conductivity K. The whole system is thermally insulated from the environment. At a moment t = 0 the temperature difference between the two chunks of metal equals $(\Delta T)_0$. Assuming the heat capacity of the rod to be negligible, find the temperature difference between the chunks as a function of time.

HLP Answers

1. $\frac{3K_0A(\theta_2-\theta_1)}{8l}$ **2.** $\frac{K\pi R_1 \quad R_2(T_H-T_C)}{L}$ **3.** (a) 1 : 4 (b) 2.9 : 1

4. (a) $ms(T_1 - T_0)$ (b) $\frac{In \quad 2}{k}$ **5.** 6.56×10^4 sec

6. (a) 180 W (b) 18 W (c) 9 W (d) $\frac{100}{19}$ s **7.** $\frac{K\pi(R_2^2-R_1^2) \quad (T_2-T_1)}{l}$

8. $\frac{2\pi\ell k}{\ell n(r_2/r_1)}$ **9.** $\frac{4\pi k r_1 r_2}{(r_2-r_1)}$ **10.** $t = mL\ln\left(\frac{r_2}{r_1}\right) \Big/ 100\pi k\ell$

11. $\frac{mCL}{KS}\ell \quad n \left(\frac{T_0-T_1}{T_0-T_2}\right)$ **12.** 887 J **13.** 539 W, 375 W, 164 W **14.** 92.2 W

15. 9 kW **16.** $T(x) = T_1\left(\frac{T_2}{T_1}\right)^{x/\ell}$; $q = (\alpha/\ell)\ \ell n\left(\frac{T_1}{T_2}\right)$

17. $\Delta T = (\Delta T)_0\, e^{-\alpha t}$, where $\alpha = (1/C_1 + 1/C_2)\, SK/\ell$